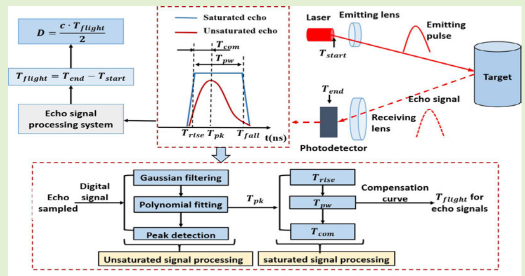
A Sampling-Based Pulsewidth-Compensated Timing Method Independent of Distance and Reflectivity for LiDAR Application

Fahu Xu and Dayong Qiao

***Abstract*—Light detection and ranging (LiDAR) with a high-ranging accuracy and large detection range serves as a critical component in the field of optical measurement. In this article, we specifically propose a sampling-based pulsewidth-compensated timing method for LiDAR ranging, which reveals the pulsewidth compensation value independent of distance and reflectivity. Characteristics of echo signals are analyzed to set up the relationship among the pulsewidth, compensation value, and time of flight. The implementation procedure of the timing method is described in detail, and ranging parameter calibration and experiments are carried out. Distance measurement precision and accuracy within 5 cm are consistently achieved across target boards with a reflectivity of 10% and 90% over ranges from 1 to 50 m. Eventually, a MEMS-based single-beam LiDAR system works to verify the ranging ability, and the experimental results show that the sampling-based pulsewidth-compensated timing method provides a promising potential for LiDAR large dynamic ranging.**



***Index Terms*— Distance and reflectivity, light detection and ranging (LiDAR), pulsewidth compensation, timing method.**

I. INTRODUCTION

LIGHT detection and ranging (LiDAR) as an active optical measurement technology has been extensively used for various application scenarios, such as autonomous driving [1], robotics [2], and autonomous aerial vehicles (UAVs) [3]. A LiDAR usually emits laser pulses onto an object and detects the target's distance based on the direct time-of-flight (dToF) technology [4], in which the round-trip time interval between the emitted pulse and received pulse is measured. Since the ns-scale emitted and received pulses in the ToF principle, an accurate timing method is required.

There typically exist two common timing methods for recording the time of flight in the dToF technology. One timing method is based on the high-speed analog-to-digital converter (ADC), which samples the analog emitted signal and received signal, and then the digitalized full-waveform signals can be reconstructed [5]. Some digital signal-processing algorithms are usually implemented on the full-waveform signals to extract the time of flight, such as peak discrimination [6], leading edge discrimination [7], center of gravity discrimination [8], and inflection discrimination [7]. However, digital signal-processing algorithms do not work well with the saturated echo signals brought by close-range highly reflective objects, due to the saturated signals have totally lost the characteristics of the Gaussian pulse [9]. The other timing method is based on the time-to-digital converter (TDC), which converts the rising edge time or falling edge time of the analog trigger signals into digital information [10]. Then, the time of flight can be calculated by subtracting the start trigger time from the end trigger time. Typically, a constant fraction discriminator (CFD) is used in the TDC method to reduce the timing walk error due to the variation of echo amplitude [11], [12]. Nevertheless, the timing walk error is difficult to eliminate in the TDC method, especially on long-range low-reflectivity objects [13].

In this study, we propose a sampling-based pulsewidth-compensated timing method for the LiDAR ranging system. This timing method employs a high-speed ADC to sample analog echo signals and reconstruct digital full waveforms.

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The characteristics of echo signals are analyzed in detail, including the rising edge time, falling edge time, pulsewidth, and peak time. For the unsaturated signal, the accurate time of flight can be obtained by employing a digital signal-processing algorithm. For the saturated signal, the accurate time of flight is determined by the established relationship among pulsewidth, compensation value, and time of flight. Based on the compensation curve of pulse widths, the accurate time of flight of echo signals with different saturation levels is obtained. The entire echo processing and parameter calibration procedure is specifically presented, which proves that the pulsewidth compensation value is independent of distance and reflectivity. To verify the feasibility of this pulsewidth-compensated timing method, a MEMS-based LiDAR system is set up, and the ranging experiments under different conditions are carried out. Eventually, the experimental results show that a favorable ranging accuracy is achieved with the pulsewidth-compensated timing method. This study provides a reasonable potential to achieve a dynamic distance measurement for the LiDAR system.

The rest of this article is organized as follows. Section II gives a detailed explanation of the sampling-based pulsewidth-compensated timing method and the entire processing procedure of the echo signal. Section III presents the setup of the MEMS LiDAR system and timing method simulation. In Section IV, the parameter calibration and ranging experiments are carried out to verify the feasibility of this pulsewidth-compensated timing method. In Sections V and VI, the discussion and conclusion are presented.

II. Methodology

A. Echo Characteristics of the dToF Ranging Method

The principle of the dToF ranging method is to time the duration that it takes for a laser pulse to travel from the laser emitter to the photodetector. The target distance is then calculated based on the flight time of the laser pulse, which can be indicated as

$$D = \frac{c \cdot T_{flight}}{2} = \frac{c\,(T_{end} - T_{start})}{2} \tag{1}$$

where c is the speed of light, T_{flight} is the time difference in light propagation, T_{start} is the start time of the laser flight, T_{end} is the end time of the laser flight, and D is the target distance.

In laser dToF distance measurement, as shown in Fig. 1, the moment that a laser emission system sends out a laser pulse is defined as T_{start} of laser flight. After traveling a specific distance, the pulse hits the target and reflects off it and is then detected by the photodetector. The echo signal-processing system amplifies and filters the echo signal. The timing system records the signal-processing time and provides T_{end} of laser flight. Eventually, T_{flight} is obtained by subtracting T_{start} from T_{end}, and D can be obtained based on the laser flight speed.

For dToF ranging, the ranging accuracy is primarily determined by laser pulse timing precision. Hence, it is crucial to analyze the characteristics of laser pulses. Typically, the width of the laser pulse is in the nanosecond range, and its waveform closely resembles a Gaussian pulse [14], which can be defined as

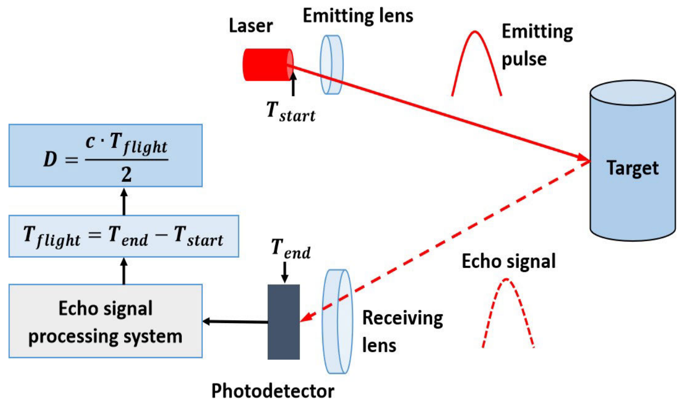

Fig. 1. Principle of the dToF ranging method.

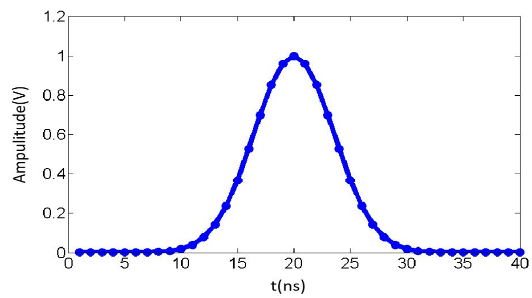

Fig. 2. Simulation of a Gaussian pulse.

$$S(t) = A_s \cdot \exp\left(-\left(\frac{t - B_s}{C_s}\right)^2\right) \tag{2}$$

where $S(t)$ is the function of the Gaussian pulse, t is the time, A_s is the pulse amplitude, B_s is the position of pulse peak, and C_s is the pulsewidth coefficient, which is proportional to the full-width at half-maximum (FWHM). For Gaussian pulse simulation and analysis, A_s is defined as 1 V, B_s is defined as 20 ns, and C_s is defined as 5 ns. Fig. 2 shows the simulated waveform of a Gaussian pulse.

The dToF ranging method derives laser time-of-flight by analyzing distinct pulse characteristics: peak amplitude, rising/falling edge timing, and pulsewidth. Consequently, rigorous characterization of the echo waveform is essential, as its features directly depend on detectable optical power. Given predetermined LiDAR transceiver design parameters, the received optical power is proportional to the ratio of the target reflectivity to the square of the target distance [15]

$$P_r \propto \frac{\rho}{D^2} \tag{3}$$

where P_r is the amount of received optical power and ρ is the target reflectivity [16]. The target reflectivity is an inherent property of a natural object and typically falls between 0.1 and 0.9. Since the LiDAR system usually needs to detect distances over 100 m, the dynamic range of the laser echo signal can vary by a factor of more than 10 000 [17], [18].

To further analyze the characteristics of the laser echo signal, laser echoes under different detection scenarios can

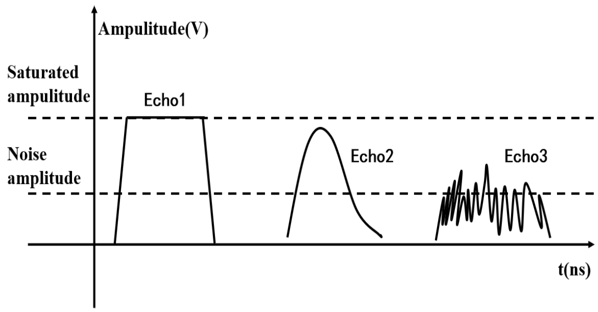

Fig. 3. Laser echoes under different detection scenarios.

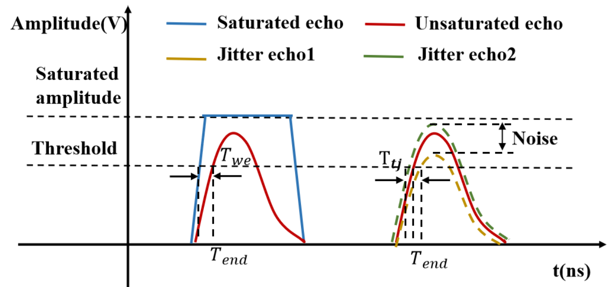

Fig. 4. Principle of the TDC timing method.

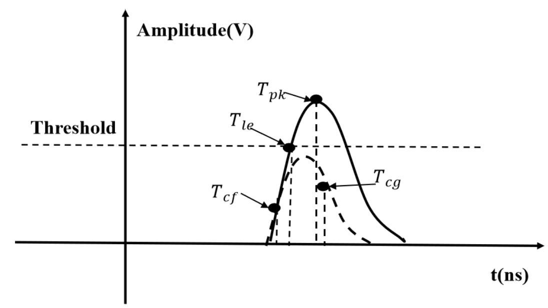

Fig. 5. Principle of the ADC timing method.

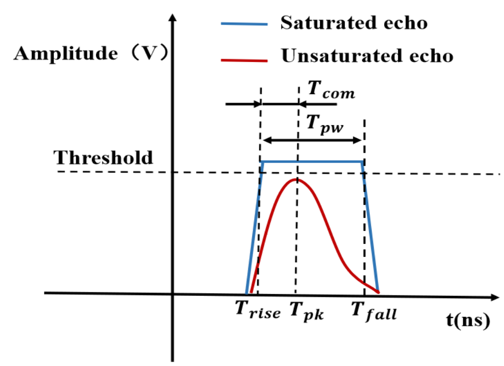

Fig. 6. Principle of the pulsewidth-compensated timing method.

be divided into three categories, as shown in Fig. 3. When measuring a close-range target with high reflectivity, the strength of Echo1 surpasses the linear amplification capacity of the circuit, resulting in a saturated signal with a clipped amplitude [19]. For a medium-range target with moderate reflectivity, the strength of Echo2 is within the linear amplification range of the circuit, not clipped, and has an amplitude above the noise level, indicating an unsaturated signal. However, for a long-range target with low reflectivity, Echo3 is nearly obscured by noise, making detection challenging.

By analyzing the echo signal characteristics under different scenarios, it can be observed that unsaturated signals exhibit the characteristics of Gaussian pulses, with measurable peak amplitude, rising edge, falling edge, and pulsewidth. Saturated signals display clipping distortion, lacking peak information, and the overall signal experiences broadening. To reliably measure targets exceeding 100-m distances with reflectivities spanning 0.1–0.9, the LiDAR system must extract accurate laser flight time from both unsaturated and saturated echo signals.

B. Sampling-Based Pulsewidth-Compensated Timing Method

There are currently two primary methods for timing the laser echo signals, the TDC timing method and ADC timing method, as shown in Figs. 4 and 5. The principle of the TDC timing method involves using a comparator to trigger the rising edges of both laser-transmitted and echo signals, thereby generating T_{end}. The TDC timing method usually requires the echo receiving circuit with high bandwidth and stability to reduce timing jitter error T_{jt} and wander error T_{we} and has an advantage in timing saturated echo signals. The principle of the ADC timing method utilizes a high-speed ADC to convert the echo signal into a digital signal, and digital signal-processing algorithms are implemented to obtain T_{end}, such as leading edge discrimination (T_{le}), peak detection (T_{pk}), centroid method (T_{cg}), and constant fraction discrimination (T_{cf}). The timing accuracy of the ADC timing method depends on the ADC's sampling precision and has an advantage in timing unsaturated echo signals.

Leveraging complementary advantages of TDC and ADC methods, this article proposes a pulsewidth-compensated timing technique. This timing method employs high-speed ADC acquisition for full waveform digitization, applying distinct signal-processing pathways for unsaturated and saturated echo signals. For unsaturated echo signals, digital signal-processing algorithms are used to obtain the peak time T_{pk} of the echo signal, which is taken as T_{end}. When the amplitude of the echo signal reaches the maximum sampling value of the ADC, the echo signal is defined as saturated. For saturated echo signals, the rising edge time T_{rise}, falling edge time T_{fall}, and pulsewidth T_{pw} are extracted through a threshold trigger, which is set to the maximum sampling value of the ADC. Then, based on T_{pk} of the unsaturated echo signal for the same target, the timing compensation value T_{com} for the saturated echo signal at that pulsewidth is calculated. T_{end} for the saturated echo signal is obtained by adding T_{com} to T_{rise}. Fig. 6 illustrates the principle of the sampling-based pulsewidth-compensated timing method

$$T_{end} = T_{pk} \tag{4}$$

$$T_{\text{com}} = T_{pk} - T_{rise} \tag{5}$$

$$T_{end} = T_{rise} + T_{\text{com}}. \tag{6}$$

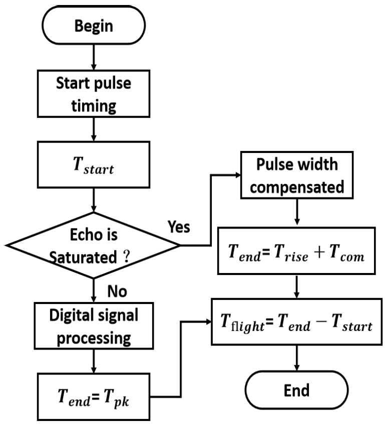

Fig. 7. Workflow of the pulsewidth-compensated timing method.

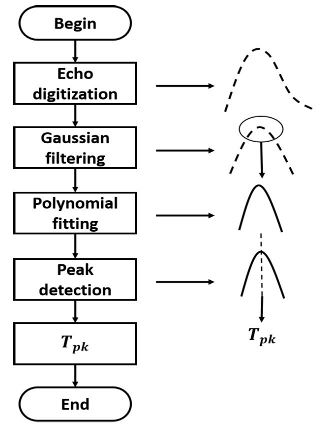

Fig. 8. Derivation process of T_{pk}.

In the pulsewidth-compensated timing method, T_{end} of the unsaturated signal is calculated as (4), T_{com} is calculated as (5), and T_{end} of the saturated signal is calculated as (6). It should be noted that T_{pw} of the echo signal corresponds one to one with T_{com}, and a relationship curve between T_{pw} and T_{com} can be established after calibration. T_{pw} of the echo signal represents the identical intensity, and echo signals with the same intensity may appear on targets that are either close-range and low-reflectivity or far-range and high-reflectivity. Therefore, the correspondence between T_{pw} and T_{com} is independent of the target's distance and reflectivity.

Fig. 7 details the sampling-based pulsewidth-compensated timing workflow. First, T_{start} is extracted from the start pulse, which is a saturated signal with a stabilized rising edge. Then, determine the echo signal saturation by applying an amplitude threshold criterion. When the echo signal is unsaturated, T_{pk} is obtained through digital signal-processing algorithms and employed as the laser flight end time T_{end}. When the echo signal is saturated, T_{end} is calculated by adding T_{rise} to T_{com}. The laser flight time T_{flight} is obtained by subtracting T_{start} from T_{end}.

Within the sampling-based pulsewidth-compensated timing framework, T_{pk} of the unsaturated echo signal is considered as the accurate timing moment and employed to obtain T_{com} of the saturated echo signal. Therefore, T_{pk} critically governs the timing accuracy of the timing method, with its derivation process detailed in Fig. 8. First, a high-speed ADC converts the analog echo signal into a discrete digital signal, which is then subjected to Gaussian filtering [20]. The discrete Gaussian signal is correlated with the echo data to reduce the impact of system noise. Subsequently, the least-squares method is used to perform polynomial fitting on a portion of the filtered echo data, enhancing timing accuracy [21]. Finally, peak detection is applied to the fit echo data to extract T_{pk} of the unsaturated signal.

Gaussian filtering of unsaturated echo signals reduces timing jitter errors induced by system noise, implemented as (7). Here, E_r is the digital echo signal, E_f is the echo signal processed by Gaussian filtering, f_d is the discrete Gaussian signal related to A_s, B_s, and C_s. n is the number of data in E_r, m is the number of data in f_d, and i and j are the position of data

$$E_f\left(j\right)=\sum_{i=1}^{m}E_r\left(i+j-1\right)f_d\left(i\right)\quad j=1,2,3,\ldots n-m. \tag{7}$$

Polynomial fitting is indicated as (8) to improve the timing accuracy of the echo signal. Here, E_p is the echo data required for fitting, typically selected from the partial peak values of E_f based on the waveform characteristics of the echo signal. Y_t is the data after fitting, a_0 is the coefficient of the quadratic term, a_1 is the coefficient of the linear term, and a_2 is the coefficient of the constant term, and k_1, k_2, k_3, and k_n are the positions of data in E_p

$$E_p\left(t\right)\approx Y_t=a_0t^2+a_1t+a_2\quad t=k_1,k_2,k_3,\ldots,k_n \tag{8}$$

where a_0, a_1, and a_2 are obtained based on the principle of the least-squares method. The least-squares method aims to minimize the total squared difference between Y_t and E_p as shown in (9) and (10). Here, δ represents the deviation between the original data and the fit data, ER^2 represents the sum of the squared deviations between the original data and the fit data

$$\min\sum_{t=k_1}^{k_n}\delta\left(t\right)^2=\sum_{t=k_1}^{k_n}\left(Y_t-E_p\left(t\right)\right)^2\quad t=k_1,k_2,k_3,\ldots k_n \tag{9}$$

$$\text{ER}^2=\sum_{t=k_1}^{k_n}\left(E_p\left(t\right)-\left(a_0t^2+a_1t+a_2\right)\right)^2$$
$$t=k_1,k_2,k_3,\ldots k_n. \tag{10}$$

To minimize the sum of squared deviations between the fit data and the original data, the partial derivatives of the polynomial fitting coefficients a_0, a_1, and a_2 are sequentially calculated with respect to (10), and expressed in matrix form, resulting in the Vandemonde determinant as shown in (11). Subsequently, according to the rules of matrix operations, the

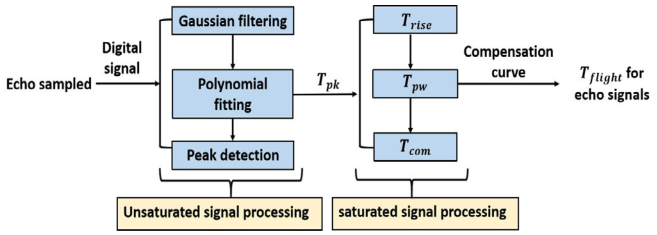

Fig. 9. Implementation procedure of the pulsewidth-compensated timing method.

polynomial fitting coefficients a_0, a_1, and a_2 are obtained

$$\begin{bmatrix} 1 & k_1 & k_1^2 \\ 1 & k_2 & k_2^2 \\ \vdots & \vdots & \vdots \\ 1 & k_n & k_n^2 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} E_p(k_1) \\ E_p(k_2) \\ \vdots \\ E_p(k_n) \end{bmatrix}. \tag{11}$$

Peak discrimination is performed on the fit echo, and the peak time T_{pk} is obtained by the peak calculation method of the quadratic curve as shown in the following equation:

$$T_{pk} = \frac{a_1}{-2a_0}. \tag{12}$$

As outlined in the above analysis, the sampling-based pulsewidth-compensated timing method differentiates between unsaturated and saturated signals for digital signal processing to achieve precise timing as shown in Fig. 9. For unsaturated signals, T_{pk} is identified through digital processing and designated as the accurate T_{end}. In the case of saturated signals, T_{pw} is first obtained from a threshold trigger and then T_{com} is calculated from a system-calibrated pulsewidth compensation curve, resulting in the accurate T_{end}. Compared to the TDC timing method, the sampling-based pulsewidth-compensated timing method minimizes timing jitter errors by processing unsaturated signals digitally. Moreover, unlike the ADC timing method, the sampling-based pulsewidth-compensated timing method reduces timing wander errors by accurately calculating T_{end} for saturated signals through pulsewidth compensation.

III. System Setup and Method Simulation

A. MEMS-Based LiDAR System Setup

To validate the feasibility of the sampling-based pulsewidth-compensated timing method, a semi-coaxial MEMS-based LiDAR ranging system has been proposed in our early study [22], [23], [24], [25]. Two synchronous MEMS mirror modules are employed in the LiDAR system to form a semi-coaxial optical path. The MEMS mirrors have a diameter of 3 mm, resonant frequency of 1156 Hz, and optical amplitude of 60°. Fig. 10 presents the physical implementation of both the MEMS mirror module and the integrated LiDAR system.

The entire ranging workflow of the MEMS LiDAR system is presented in Fig. 11. An OSRAM pulsed laser diode with a peak power of 120 W and a wavelength of 905 nm is employed as the laser emitter. A Xilinx field-programmable gate array (FPGA) is implemented to trigger laser pulses. The laser pulse first strikes the transmitting MEMS mirror and then is reflected to hit the target. The echo optical signal from

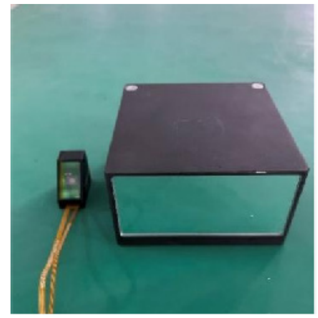

Fig. 10. Picture of the MEMS mirror module (left) and the LiDAR system (right).

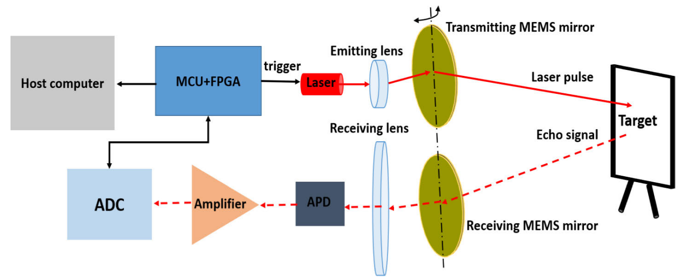

Fig. 11. Ranging workflow of the MEMS LiDAR system.

the target hits the synchronized receiving MEMS mirror and then is reflected to be received by the APD detector. A weak current signal is generated by the APD and then is converted and amplified by the amplifier. The amplifier is provided by Texas Instruments with a high gain bandwidth product of 5.5 GHz. After amplification, the echo signal is sampled by a high-speed ADC with a 1-GHz sampling rate. As a consequence, the digital echo data is transferred and processed in the FPGA and STMicroelectronics microcontroller unit (MCU). Next, an accurate time of flight is obtained by implementing the sampling-based pulsewidth-compensated timing method. Finally, an accurate measurement distance is generated, and the cloud point is transferred to the software interface.

B. Timing Method Simulation

To validate the efficacy of the pulsewidth-compensated timing method, a simulation is carried out to apply digital processing algorithms to the raw echo data. The MEMS LiDAR without scanning the MEMS mirror is employed to emit a laser pulse to the target board, and the raw echo data of the target board is captured by the ADC with a 1-GHz sampling rate. The target board has 10% reflectivity, positioned at 15 m, and an unsaturated state of the echo signal is obtained by blocking the optical path. Ten sets of unsaturated raw echo data are selected to be depicted in Fig. 12.

In Fig. 12, the horizontal axis represents the time unit, which is determined by the ADC's sampling accuracy. The ADC employed in the LiDAR system has a sampling frequency of 1 GHz, making each time unit equivalent to nanoseconds. The vertical axis represents the ADC's sampling values, which are dictated by the ADC's resolution. The ADC has an 8-bit resolution, resulting in a maximum sampling value of 255. Sampling values ranging from 0 to 255 correspond to

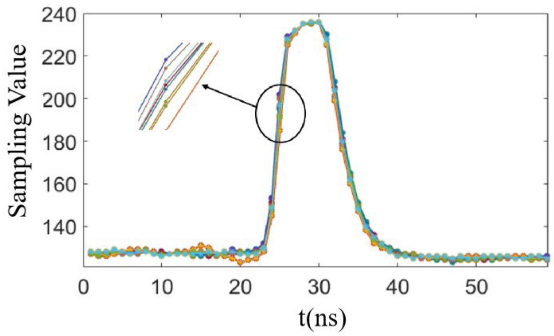

Fig. 12. Ten sets of unsaturated raw echo data.

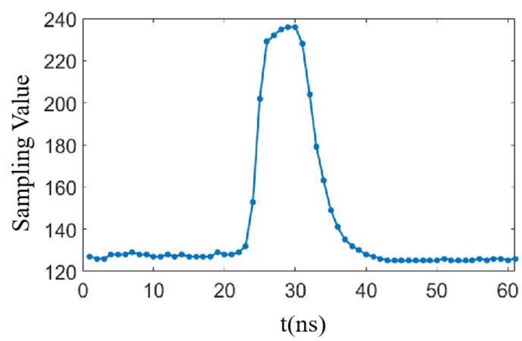

Fig. 13. One set of unsaturated raw echo data.

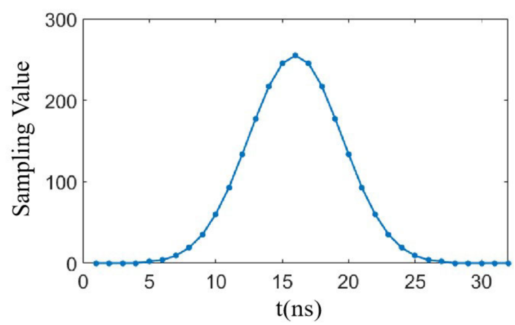

Fig. 14. Discrete Gaussian signal.

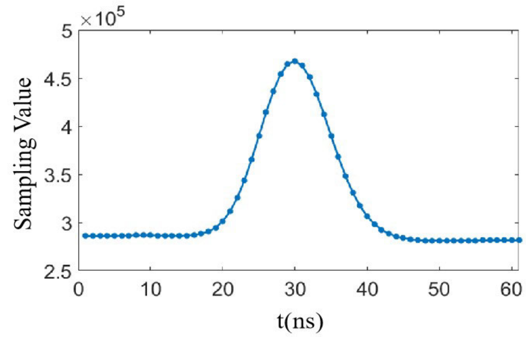

Fig. 15. Gaussian-filtered echo signal.

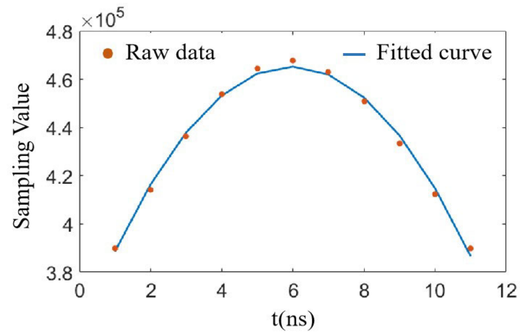

Fig. 16. Fit echo signal.

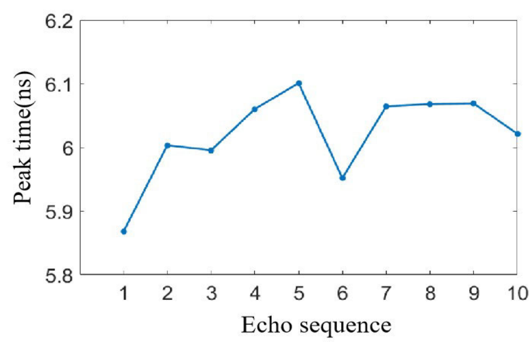

Fig. 17. T_{pk} for ten unsaturated echo signals.

input signal amplitudes from −1 to 1 V, indicating that a sampling value of 255 corresponds to an echo signal voltage of 1 V. It should be noted that the ADC's sampling values are absolute, hence the vertical axis does not carry units. Take a set of raw echo data as an example to demonstrate the processing procedure of the timing method. As shown in Fig. 13, the maximum sampling value is below 240, indicating that the echo signal is unsaturated.

The processing initiates with Gaussian filtering of raw echo data. Based on the characteristics of the echo signal, the discrete Gaussian signal is shown in Fig. 14, and the parameters of the discrete Gaussian signal f_d are set as follows: A_s is set to 255, B_s is set to 16 ns, C_s is set to 5 ns, and the number of data points is set to 32. The raw echo data after Gaussian filtering is shown in Fig. 15. The vertical axis in the following figures represents absolute values, hence it does not include units.

Next, a quadratic polynomial fitting is performed on the raw echo data after Gaussian filtering, as shown in Fig. 16. 11 data points are selected as the fitting data E_p from the peak part of the filtered echo. Finally, peak discrimination is conducted on the fitting curve to obtain the peak time T_{pk}.

Implementing the pulsewidth-compensated timing method on the ten echo datasets yields peak time measurements T_{pk} as visualized in Fig. 17. As can be seen, the T_{pk} values of the 10 echo signals are approximately 6 ns, with a maximum error jitter not exceeding 0.3 ns. It should be noted that the peak times T_{pk} are derived from the fit echo data and represent relative timing moments. The horizontal axis in Fig. 17 represents the echo sequence; thus, it does not have units.

Without obstructing the optical path, saturated echo signals from the low reflectivity target plate at 15 m are acquired, and ten sets of saturated raw echo data are selected for analysis.

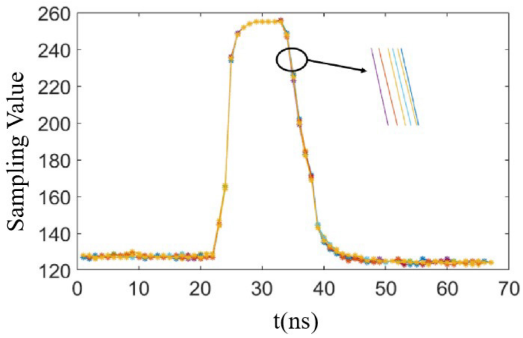

Fig. 18. Ten sets of saturated raw echo data.

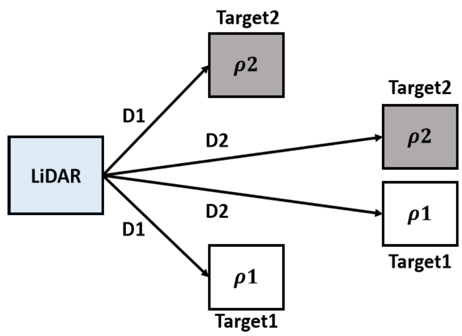

Fig. 19. Acquiring T_{pw} and T_{com} at different distances and reflectivities.

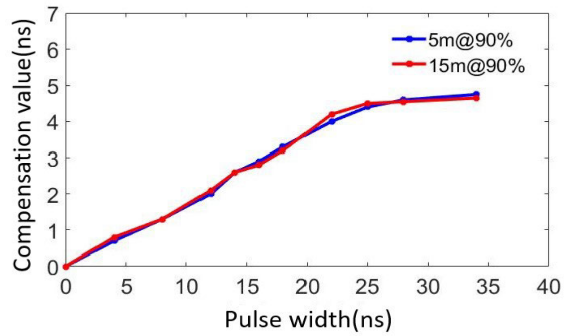

Fig. 20. Relationship between T_{pw} and T_{com} at 90% reflectivity.

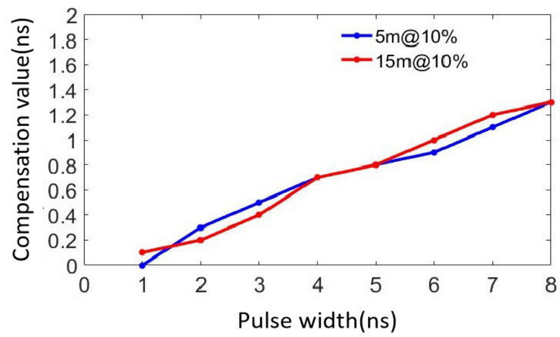

Fig. 21. Relationship between T_{pw} and T_{com} at 10% reflectivity.

As shown in Fig. 18, the ADC sampling value reaches the maximum of 255, indicating that the echo signal is in a saturated state. When the threshold is set to 255, the saturated pulsewidth is approximately 7 ns, with the jitter of T_{pw} and T_{rise} both within 0.3 ns.

It should be noted that Figs. 12–18 serve only to illustrate the detailed implementation process of the timing method and do not contain complete data. The simulation results demonstrate that accurate T_{pk} for unsaturated signals can be obtained by implementing digital signal-processing algorithms and saturated signals have stable T_{pw} and T_{rise}, which establishes a robust foundation for validating the feasibility of the sampling-based pulsewidth-compensated timing method.

IV. Experiment

A. Pulsewidth Calibration

According to the principle of the pulsewidth-compensated timing method, it is necessary to establish a bijective mapping between T_{pw} and T_{com} of saturated signals. The designed MEMS-based LiDAR system is used to detect targets at different distances and reflectivities, acquiring T_{pw} and T_{com} under various conditions as shown in Fig. 19.

Here, $\rho 1$ is 90%, $\rho 2$ is 10%, D1 is 5 m, and D2 is 15 m. At first, target1 with the reflectivity of $\rho 1$ at D1 is detected, and T_{pk} of the unsaturated signal is obtained by obstructing the optical path. Subsequently, the obstruction is gradually removed to increase the echo pulsewidth of target1 at D1, thereby acquiring different T_{pw} and the corresponding T_{com}. Next, the same procedure for target1 is conducted at D2, to obtain T_{pw} and T_{com} at a different distance. As a control, target2 with the reflectivity of $\rho 2$ is detected at D1 and D2 to obtain T_{pw} and T_{com}. Figs. 20 and 21 show the relationship between T_{pw} and T_{com} at 90% and 10% reflectivity, respectively.

As can be seen, for the target1 with 90% reflectivity, as T_{pw} varies from 0 to 35 ns, T_{com} gradually increases to approximately 5 ns. At both 5- and 15-m distances, the growth curve of T_{pw} and T_{com} is consistent, with the timing error within 0.33 ns. For target2 with 10% reflectivity, as T_{pw} varies from 0 to 8 ns, T_{com} gradually increases to approximately 1.4 ns, and the growth curve of T_{pw} and T_{com} is also consistent. Moreover, the experimental results of the two targets can serve as controls for each other and demonstrate the one-to-one correspondence between T_{pw} and T_{com}.

The calibration procedure of T_{pw} and T_{com} is shown in Fig. 22. First, detect a high reflectivity target placed at a close range. Different T_{pw} and T_{rise} of saturated signals are obtained by gradually obstructing the optical path. Second, continue obstructing the optical path at this distance until an unsaturated echo signal is acquired. Based on digital signal-processing algorithms, T_{pk} of the unsaturated signal is obtained. Third, calculate T_{com} corresponding to the saturated signal's pulsewidth at this distance by employing T_{pw} and T_{rise} from the first step and T_{pk} from the second step. Fourth, repeat the above three steps at different distances using a target with different reflectivities to verify the correctness of T_{com}. Finally, the relationship between T_{pw} and T_{com} is established, and the pulsewidth-compensated curve is plotted.

Fig. 23 demonstrates the pulsewidth-compensated curve, and experiments are conducted on targets with reflectivities of 90% and 10% within a range of 50 m to verify the correctness. The results are shown in Figs. 24 and 25. It can be observed

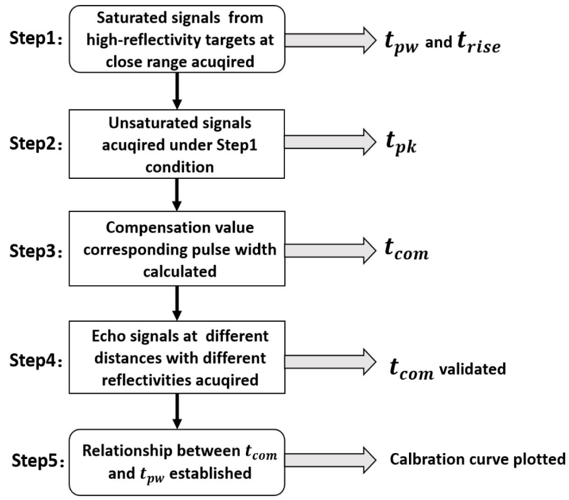

Fig. 22. Calibration procedure of T_{pw} and T_{com}.

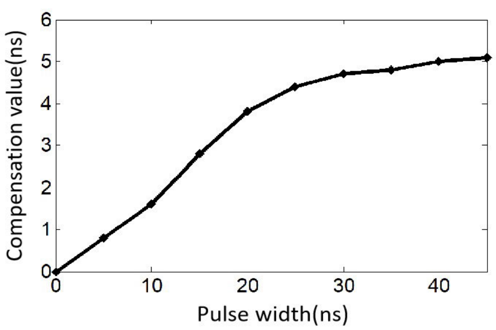

Fig. 23. Pulsewidth-compensated curve.

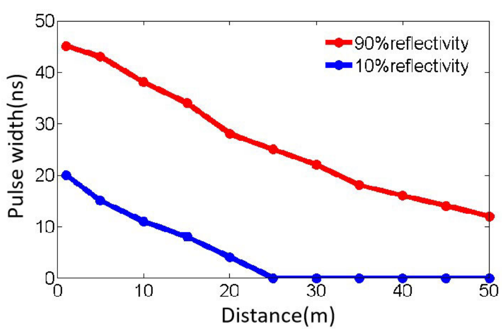

Fig. 24. Variation curve of pulsewidth.

that the echo signal from the target with 90% reflectivity is saturated within 50 m, and as the distance increases, its T_{pw} decreases from 45 to 12 ns, with the corresponding T_{com} decreasing from 5 to 2 ns. For the 10% reflectivity target, the echo signal remains unsaturated when the distance exceeds 25 m, and its T_{pw} decreases from 20 ns to 0, with T_{com} decreasing from 3.8 ns to 0. The results align with the trend of the pulsewidth-compensated curve.

Based on the above pulsewidth calibration method, a relationship curve between T_{pw} and T_{com} is established. For saturated signals, T_{end} is obtained by adding T_{rise} and T_{com}, while for unsaturated signals, T_{end} is obtained from T_{pk}. The pulsewidth compensation curve establishes a critical

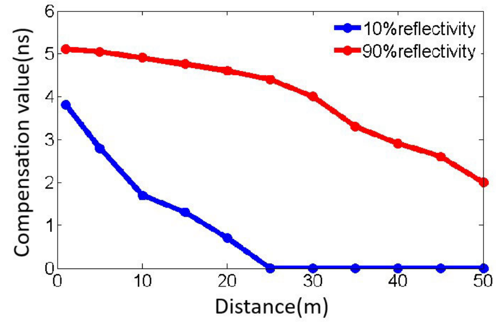

Fig. 25. Variation curve of the compensation value.

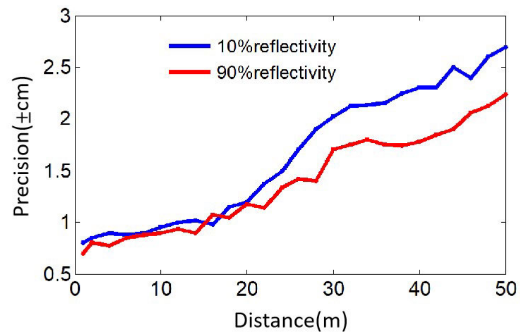

Fig. 26. Ranging precision curve.

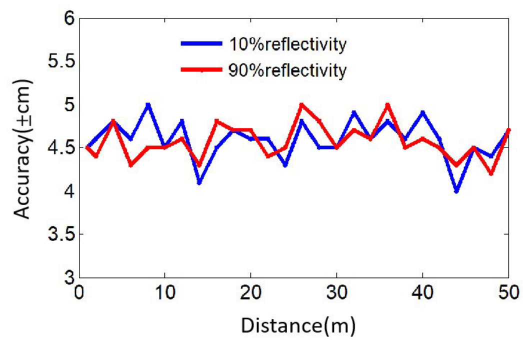

Fig. 27. Ranging accuracy curve.

foundation for implementing the sampling-based pulsewidth-compensated timing method.

B. System Ranging

Ranging experiments are carried out in which two target boards with reflectivity of 10% and 90% are detected from 1 to 50 m. The ranging precision and accuracy within 5 cm are achieved as shown in Figs. 26 and 27. The precision presents a range measurement statistical distribution and is independent of the actual range measurement. It can be calculated from the standard deviation of ranged distances as (13). The accuracy shows an error consistency of the measured distance compared with the actual distance in a defined time period, which can be defined by the following equation:

$$\mathrm{Pre} = \sqrt{\frac{1}{N-1}\sum_{i=1}^{N}\left(D_i - \bar{D}\right)^2} \tag{13}$$

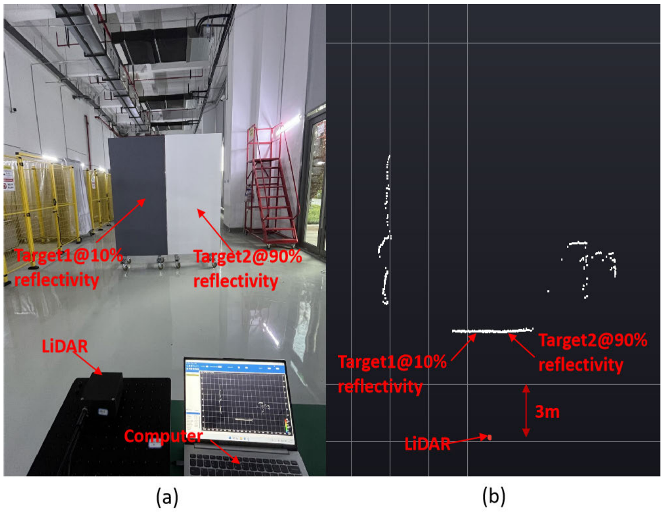

Fig. 28. (a) Scene of ranging experiment at about 6 m. (b) Cloud point of ranging experiment at about 6 m.

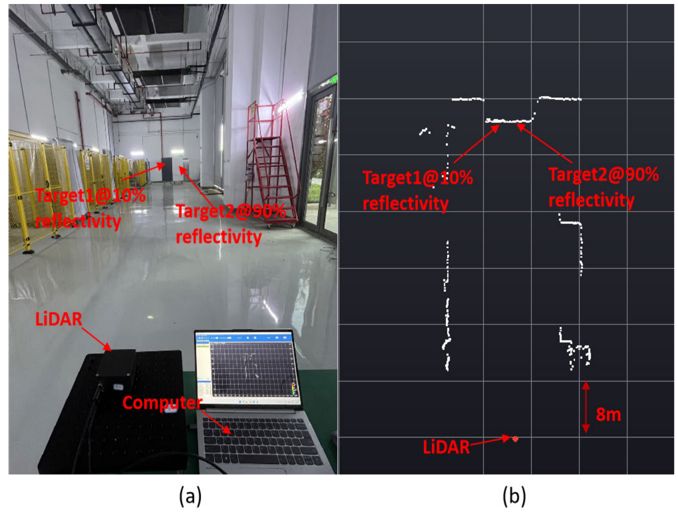

Fig. 29. (a) Scene of ranging experiment at about 45 m. (b) Cloud point of ranging experiment at about 45 m.

$$\mathrm{Acc} = \frac{1}{N}\sum_{i=1}^{N}\left(D_i - D_{\mathrm{actual}}\right). \tag{14}$$

Here, Pre is the ranging precision at a certain measured distance, N is the sampled number, D_i is the ith detected distance, $\bar{D}$ is the average of the detected distances, Acc is the ranging accuracy at a certain measured distance, and D_{actual} is the actual distance provided by a high-accuracy laser ranging finder.

To present the ranging ability vividly, the MEMS mirror starts to scan, and the semi-coaxial MEMS-based LiDAR system works. In the LiDAR imaging experiment, two target boards with reflectivity of 10% and 90% are situated side by side at about 6 and 45 m. In Fig. 28, echo signals from the two target boards are saturated, and the ranging results show the ranging consistency. In Fig. 29, echo signals from the 90% reflectivity target board are saturated, and echo signals from the 10% reflectivity target board are unsaturated. The ranging results show the ranging consistency of echo signals with different saturation levels. Based on the cloud point displayed on the software, the ranging precision and accuracy reveal a favorable consistency, whether the target is long-distance or short-distance, low-reflectivity or high-reflectivity. It should be noted that this sampling-based pulsewidth-compensated timing method provides an accurate time of flight for echo signals with different saturation levels.

V. Discussion

A direct comparative experiment was not conducted within the scope of this work; the performance of the proposed method can be robustly evaluated by analyzing the presented results. The primary objective of the compensation method is to mitigate the significant timing walk error inherent in conventional techniques like peak detection or leading-edge discrimination when processing saturated echo signals. Saturation, caused by close-range, high-reflectivity targets, leads to amplitude clipping and waveform broadening, distorting key timing features and potentially introducing large ranging errors on the order of tens of centimeters [11], [12], [13]. Our method addresses this by establishing a consistent, one-to-one correspondence between the pulsewidth of the saturated signal and a timing compensation value, effectively shifting the timing reference from a distorted feature to a stable pulsewidth metric. The most compelling evidence for the method's effectiveness is presented in Fig. 27, which shows the ranging accuracy achieved across distances from 1 to 50 m for target boards with 10% and 90% reflectivity. Critically, the accuracy remains consistently within 5 cm for both targets. Therefore, while a direct side-by-side comparison is not provided, the presented data strongly indicate that the proposed method provides a crucial enhancement, enabling precise ranging across a large dynamic range of target reflectivities and distances by effectively compensating for saturation-induced errors.

Although the signal-processing components of the pulsewidth-compensated timing method are optimized based on a Gaussian waveform model, the method is not inherently limited to Gaussian waveforms. The core concept of the method is to utilize the one-to-one correspondence between the pulsewidth and the timing compensation value to correct the time-of-flight for saturated signals. The universality of this approach is grounded in the fundamental physical phenomenon of pulse broadening due to saturation, which is a common occurrence regardless of the initial pulse shape. The compensation curve is precisely derived through experimental calibration to characterize this universal phenomenon. Future work will include validating the robustness of this method across a wider variety of pulse waveforms.

The proposed method does introduce signal-processing delay, primarily stemming from digital logic operations including high-speed ADC sampling, Gaussian filtering, polynomial fitting, and peak/pulsewidth extraction. Based on the FPGA implementation, the total latency of the entire digital signal-processing chain is calculated to be approximately 1.35 μs [23]. For the MEMS LiDAR scanning system applied in this study, which operates at a single-point measurement frequency on the order of kHz, this delay is acceptable and does not constitute a system bottleneck. Because it is significantly smaller than both the time-of-flight of the laser pulse over the maximum measurement range of 50 m and the stabilization time of the MEMS mirror scanning. Meanwhile, this delay is a fixed value, which can be compensated for during system calibration, thus ensuring it does not impact ranging accuracy.

VI. Conclusion

In this article, the characteristics of laser echo signals are mainly studied, including the pulsewidth, rising edge time, as well as falling edge time and peak time. A sampling-based pulsewidth-compensated timing method independent of the distance and reflectivity is proposed for the LiDAR ranging, in which the relationship between the pulsewidth, compensation value, and time of flight is set up. The calibration procedure of the compensation parameters is also specifically described. The experiments on the ranging precision and accuracy are carried out with different reflectivity target boards at different distances. The experimental results show that a 5-cm ranging precision and accuracy is obtained under echo signals with different saturation levels. Experimental validation using a MEMS-based single-beam LiDAR system demonstrates that the pulsewidth-compensated timing method enables large dynamic range measurement, with sampling-based calibration showing significant potential for extended applications.

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